Jack Hogan

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Computer science and artificial intelligence engineer with experience across software engineering, AI/ML research, and technical leadership. Focused on deep reinforcement learning and systems that let models reason, adapt, and learn complex behaviors, with applications in robotics, vision, and low-data learning.

Education

Purdue University Honors College Fall 2022 — Spring 2026

Computer Science B.S., Artificial Intelligence B.S.; Minor in Mathematics; Graduated Spring 2026 *West Lafayette, IN*

- Relevant Coursework: Data Structures, Graduate Machine Learning, Artificial Intelligence, Analysis of Algorithms, Abstract Algebra
- Study Abroad: University College Dublin, Spring 2024

Work Experience

Era Labs August 2026 — Present

Founding Embedded Systems Engineer *San Francisco, CA*

- Designed and implemented a patent-pending firmware architecture supporting dynamic peripheral registration and bytecode execution for ~100ms idea iteration.
- Designing and fabricating circuits for new product lines with microamp-scale power draws and efficient layouts.
- Designed a cohesive software stack for registering, finding, and collaborating with Beacons, a hardware/software solution for fostering real-time social connections at events, using traditional and LLM-based machine learning techniques.

Peraton Labs May 2025 — May 2026

Machine Learning Researcher *Silver Spring, MD*

- Created a new pipeline for concurrently testing different GNN architectures, speeding up testing by 2-3x.
- Investigated and implemented multiple bleeding-edge GNN architectures, improving task accuracy by 50%+.
- Developed and presented a technical report on a novel GNN architecture, demonstrating significant performance gains and guiding next-phase research.

Leidos Innovation Center May 2024 — August 2024

Software Engineering Intern, High-Performance Systems *Arlington, VA*

- Created and refined GPU-based signal processing algorithms in CUDA, Rust, and Python for real-time applications, improving performance by 400%.
- Developed a real-time machine learning system to lock onto hidden radar targets using PyTorch with 85%+ accuracy.

Peraton May 2023 — August 2023

Software Engineering Intern *Chantilly, VA*

- Automated document processing systems using AI-based parsing systems to increase processing efficiency.
- Authored and presented a whitepaper on systems and processes developed.
- Managed daily intern standups and kept tasks on track.
- Created data ingest and management dashboards for U.S. contractors.

Alluja LLC 2022 — Present

Founder *Chantilly, VA*

- Creator of multiple products including mobile apps (SwiftUI), libraries (Swift, Rust), backend servers (Rust, Python), and frontends (Svelte, Leptos) impacting 5,000+ users.
- Komori, a Discord Siri Shortcuts integration, with paid downloads on the App Store.
- Alluja WebSockets, a WebSocket testing client for macOS, with 100+ downloads.

Research & Leadership

Purdue Hackers Jan 2024 — May 2026

Engineering Division Lead & Interim President *West Lafayette, IN*

- Oversaw all engineering projects for Purdue Hackers, a 2000+ member student organization focused on innovation and rapid prototyping.
- Managed a team of 10+ officers, balancing budget and time constraints while delivering multiple large-scale technical initiatives.
- Designed and maintained infrastructure for the Passports initiative, including authentication and management systems in Rust and TypeScript, supporting hundreds of active users.
- Led development of The Sign, a meter-tall Conway glider installation representing the club's logo; project was accepted to Hackaday and drew 1,000+ online views.
- Spearheaded the Beacons project in collaboration with Era Labs, deploying early prototypes at student events.
- Organized and ran major events, including a 200+ person callout, hack nights, and technical workshops.

Detecting Source Code Plagiarism in Submitted Assignments Spring 2023 — Fall 2025

Lead Researcher *West Lafayette, IN*

- Selected by Purdue faculty to lead an independent research project addressing plagiarism detection in the Computer Science program.
- Designed a language-agnostic detection system using Abstract Syntax Trees, parsing techniques, and machine learning to identify similarity beyond surface-level code patterns.
- Conducted large-scale testing across multiple courses, targeting 1,000+ student submissions annually to improve fairness and academic integrity.

Skills

- **Programming Languages**: Python, Rust, C, C++, C#, Java, Kotlin, Swift, TypeScript, JavaScript, SQL
- **Technologies**: PyTorch, NumPy, SciPy, Pandas, CUDA, CUTLASS, Burn, SvelteKit, Astro, React, Leptos, Actix Web, SwiftUI, Jetpack Compose, GStreamer
- **Platforms**: EC2, ECS, S3, Docker, Docker Compose, MongoDB